

UNIT #2

Exploring Derivatives of Polynomial, Radical and Rational Functions

Examples: State the derivatives of each function

a) $f(x) = x^{100}$

b) $f(x) = \pi$

c) $f(x) = -7.39$

d) $y = 9x^5$

e) $f(x) = (2x)^4$

f) $y = (x+1)^2$

g) $f(x) = x^2(x^2 + 3)$

Examples: Differentiate

a) $f(x) = (1 - 2x)^3$

b) $f(x) = (x + 2)(x^2 - 1)$

c) $s = \frac{3t^5 - 2t^2 + 5t^{-1}}{t^2}$

d) $u = \sqrt{\frac{2}{x}} + \sqrt{\frac{x}{3}}$

1. Differentiate the following functions

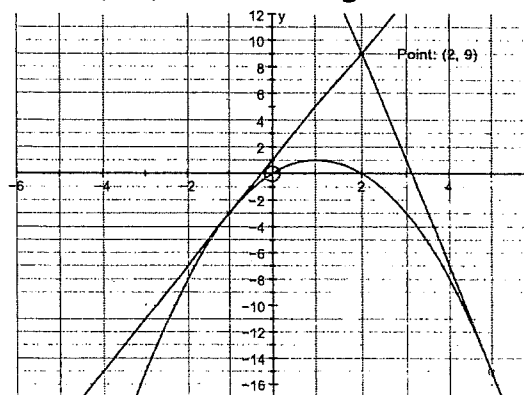
a) $f(x) = 3\sqrt{x} - \frac{1}{\sqrt{x}}$

b) $y = \sqrt{x}(x^3 - x^{\frac{1}{2}} + 5)$

c) $f(x) = \frac{2x^5 + 5x^2 + x}{3x^2}$

2. Find an equation of the tangent line to the curve $y = x^4 + 2x^2 - x$ at point $(1, 2)$.

3. Find the equations of both lines that pass through the point $P(2, 9)$ and are tangent to the parabola $y = 2x - x^2$. [$y = 4x + 1$, $y = -8x + 25$]



PROBLEM SOLVING

- 1) Find the coordinates of the point on the curve, $y = 3x^2 - 4x$, where the tangent line is parallel to the line, $y = 8x$. (2,4)
- 2) Show that there are two tangents to the curve, $y = x^5 - 2x$, that have a slope of $\frac{-3}{4}$.
- 3) Find the point on the curve, $y = x\sqrt{x}$, where the tangent line is parallel to the line, $6x - y = 4$. (16,64)
- 4) **Definition: The normal line to a curve at a point is perpendicular to the tangent at that point.**

Find the equation of the normal line to the curve, $y = -x^2 + 5x$, that has a slope of -2 . $(32x + 16y - 171 = 0)$

- 5) At what point does the curve, $y = x^3 + 3x^2 - 24x + 1$, have a horizontal tangent line? $(-4, 81), (2, -27)$
- 6) A 1000 litre tank loses water, so that after t days, the remaining volume in litres is $V(t) = 1000\left(1 - \frac{t}{10}\right)^2$ for $0 \leq t \leq 10$. How rapidly is water being lost when the tank is half full? $(100\sqrt{2} \text{ litres/day})$

Ex. 1: Differentiate: $h(x) = \sqrt{x}(1+x)$

Ex. 2: Find the equation of the tangent line to the curve

$$f(x) = (\sqrt{x} + 2)(1 - 2\sqrt{x} + 3x) \text{ at } x = 1.$$

Ex. 1: Differentiate $h(x) = \frac{x^2+2x-3}{x^3+1}$.

Ex. 2: Find $\frac{dy}{dx}$ if $y = \frac{\sqrt{x}}{1+2x}$.

Jazz Day Review

1. Find the derivatives of each of the following functions

a) $f(x) = (x^5 + 3x^3 - 4x^2 + 1)^5 [f'(x) = 5x(5x^3 + 9x - 8)(x^5 + 3x^3 - 4x^2 + 1)^4]$

b) $f(x) = \frac{1}{x^3+5} [f'(x) = -3x^2(x^3 + 5)^{-2}]$

c) $f(x) = \frac{6}{(x^3+2x)^2} [f'(x) = -12(3x^2 + 2)(x^3 + 2x)^{-3}]$

d) $f(x) = (5x - 8)(3x^2 + 12) [f'(x) = 45x^2 - 48x + 60]$

e) $f(x) = (x^2 + 1)(x^5 - 3x + 1)^3 [f'(x) = (x^5 - 3x + 1)^2(17x^6 + 15x^4 - 15x^2 + 2x - 9)]$

f) $f(x) = (7x - 5)^4(x^3 + 5x)^3 [f'(x) = (7x - 5)^3(x^3 + 5x)^2(9x^3 - 45x^2 + 245x - 75)]$

g) $f(x) = \frac{(2x-5)^4}{(8x^2-5)^3} [f'(x) = -8(2x - 5)^3(8x^2 - 5)^{-4}(4x^2 - 30x + 5)]$

h) $f(x) = \sqrt[3]{\frac{x^2+3x}{5x-7}} [f'(x) = (x^2 + 3x)^{-\frac{2}{3}}(5x - 7)^{-\frac{4}{3}}\left(\frac{5x^2}{3} - \frac{14x}{3} - 7\right)]$

Challengers

i) $f(x) = [5x^4 - (x^3 + 8x)^{-3}]^3$

j) $f(x) = \sqrt[3]{2x + \sqrt{x^2 + 3}}$

2. A table of values for f, g, f' , and g' is given

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
1	7	1	6	3
2	1	8	4	1
3	3	2	-3	-9

a) If $h(x) = f(x)g(x)$, find $h'(2)$ [33]

d) If $h(x) = g(g(x))$, find $h'(1)$ [9]

b) If $h(x) = f(g(x))$, find $h'(3)$ [-36]

c) If $h(x) = 2f(x) - 3g(x)$, find $h'(1)$ [3]

3. Find the equation of the tangent line to the curve $y = (x^3 - 10)^2$ at $x = 2$ [$y = -48x + 100$]

4. Find the equation of the tangent line to the curve $f(x) = (3x + 13)^4$ that is parallel to the line $y = -96x + 35$ [$y = -96x - 464$]

5. Find the second derivative of $f(x) = (3x - 2)^2(7x)$ [$f''(x) = 378x - 168$]

THE CHAIN RULE - Exercises

1. Differentiate each of the following:

a. $y = (3 + x^3)^5$

e. $y = \frac{1}{4x - x^2}$

b. $y = \left(1 - \frac{x}{6}\right)^7$

f. $y = \sqrt{3x - 2}$

c. $y = \frac{1}{4}(3x^2 - 2x + 8)^{24}$

g. $y = \sqrt[3]{x^2 - 9}$

d. $y = \frac{1}{(3 - 4x)^2}$

h. $y = (ax^2 + bx + c)^m$

2. Find the equation of the tangent line to the curve $y = \sqrt{7 + \sqrt{x}}$ at the point (4, 3).

3. Find $\frac{dy}{dx}$ at the indicated value of x .

a. $y = 2u^3 + 3u^2$, $u = x + \frac{1}{\sqrt{x}}$, at $x = 1$.

b. $y = \frac{1}{(1 + u^2)^2}$, $u = \sqrt{x} - 1$, at $x = 4$.

c. $y = u^5 + u^3$, $u = 4v + 3v^{-1}$, $v = 3 - x^2$ at $x = 2$.

4. Differentiate $f(x) = \frac{2x^2 - 2x + 7}{x - 3}$ by using long division to first write the function in the form $f(x) = Ax + B + \frac{C}{x - 3}$.

5. Find a formula for $\frac{dy}{dx}$ when $x = \frac{1}{2}$ if:

a. $y = f(x^3)$

b. $y = [f(x)]^3$

c. $y = (1 + f(x))^2$

6. Let $y = f(x^2 + 3x - 5)$. Find $\frac{dy}{dx}$ when $x = 1$, given that $f'(-1) = 2$.

7. Let $p = q(r(t))$ where $r(t) = (t^2 + t + 2)^{-1/3}$. If $q'\left(\frac{1}{2}\right) = 3$, find $\frac{dp}{dt}$ when $t = 2$.

8. Let $y = \sqrt{x + f(x^2 - 1)}$. Find $\frac{dy}{dx}$ when $x = 3$, given that $f(8) = 0$ and $f'(8) = 3$.

Answers

1. a. $15x^2(3+x^3)^4$ b. $-\frac{7}{6}\left(1-\frac{x}{6}\right)^6$ c. $12(3x-1)(3x^2-2x+8)^{23}$
 d. $\frac{8}{(3-4x)^3}$ e. $\frac{2(x-2)}{x^2(4-x)^2}$ f. $\frac{3}{2\sqrt{3x-2}}$
 g. $\frac{2x}{3\sqrt[3]{x^2-9}^2}$ h. $m(2ax+b)(ax^2+bx+c)^{m-1}$

2. $x - 24y + 68 = 0$

3. a. 18 b. $-\frac{1}{8}$ c. -48608

4. $2 - \frac{19}{(x-3)^2}$

5. a. $\frac{3}{4}f'\left(\frac{1}{8}\right)$ b. $3\left[f\left(\frac{1}{2}\right)\right]^2 f'\left(\frac{1}{2}\right)$ c. $2\left(1+f\left(\frac{1}{2}\right)\right)f'\left(\frac{1}{2}\right)$

6. 10

7. $-\frac{5}{16}$

8. $\frac{19\sqrt{3}}{6}$

⑨ Use the Chain Rule, in Leibniz notation, to find $\frac{dy}{dx}$ at the indicated value of x .

a. $y = 3u^2 - 5u + 2, u = x^2 - 1, x = 2$

b. $y = u^3 - 5(u^3 - 7u)^2, u = \sqrt{x}, x = 4$

c. $y = 2u^3 + 3u^2, u = x + x^{\frac{1}{2}}, x = 1$

d. $y = u(u^2 + 3)^3, u = (x + 3)^2, x = -2$

e. $y = \frac{u^3}{u+1}, u = (x^2 + 1)^3, x = 1$

f. $y = \frac{1}{(1+u^2)^2}, u = \sqrt{x} - 1, x = 4$

g. $y = u^5 + u^3, u = \frac{3}{v} - 4v, v = 3 - x^2, x = 2$

⑩ Let $y = f(x^2 + 3x - 5)$. Find $\frac{dy}{dx}$ when $x = 1$, given that $f'(-1) = 2$.

Application ⑪ Let $y = g(h(x))$ where $h(x) = \frac{x^2}{x+2}$. If $g'(\frac{9}{5}) = -2$, find $\frac{dy}{dx}$ when $x = 3$.

9. a. 52 b. 78 c. 54 d. 320 e. $\frac{9728}{27}$ f. $-\frac{1}{8}$ g. -48608
10. 10
11. $-\frac{42}{25}$

EXERCISE 2.8

B 1. Find the first and second derivatives of the given functions.

(a) $f(x) = x^5 - 4x^2 + 1$

(b) $g(x) = 7x^4 + 12x^3 - 4x + 8$

(c) $f(t) = 2t - \frac{1}{t+1}$

(d) $g(t) = \frac{4}{\sqrt{t}}$

(e) $y = (2x + 1)^8$

(f) $y = t^3 + \frac{1}{t^3}$

(g) $y = \sqrt{x^2 + 1}$

(h) $y = \frac{t}{t-1}$

2. Find the third derivative.

(a) $f(x) = 1 - 12x + 4x^2 - x^3$ (b) $f(x) = \frac{1}{x^5}$

(c) $y = \frac{3}{(4-x)^2}$

(d) $y = \sqrt{1+2x}$

3. Find the first six derivatives of the function

$y = x^5 + x^4 + x^3 + x^2 + x + 1.$

4. If $f(x) = \sqrt{1+x^3}$, find $f'''(2)$.

5. If $g(x) = \frac{1}{\sqrt{3x+4}}$, find $g'''(4)$.

4. $\frac{3}{2} - \frac{131}{405}$

5. $y^{(4)} = 120x + 24, y^{(5)} = 120, y^{(6)} = 0$
 $20x^3 + 12x^2 + 6x + 2, y''' = 60x^2 + 24x + 6,$
 $3. y' = 5x^4 + 4x^3 + 3x^2 + 2x + 1, y'' =$

(c) $y''' = \frac{(4-x)^5}{72}$ (d) $y''' = 3(1+2x)^{-\frac{7}{2}}$

2. (a) $f'''(x) = -6$ (b) $f'''(x) = -\frac{210}{x^8}$

(h) $y' = -\frac{(t-1)^2}{2} y'' = \frac{(t-1)^3}{2}$

(g) $y' = x(x^2 + 1)^{-\frac{1}{2}}, y'' = x(x^2 + 1)^{-\frac{3}{2}}$

(f) $y' = 3t^2 - 3t^{-4}, y'' = 6t - 12t^{-5}$

(e) $y' = 16(2x+1)^{-\frac{1}{2}}, y'' = 224(2x+1)^{-\frac{3}{2}}$

(d) $g'(t) = -2t^{-\frac{3}{2}}, g''(t) = 3t^{-\frac{5}{2}}$

(c) $f'(t) = 2 + \frac{(t+1)^2}{2} f''(t) = -\frac{(t+1)^3}{2}$

$f'''(x) = 84x^2 + 72x$

(b) $f'(x) = 28x^3 + 36x^2 - 4,$

1. (a) $f'(x) = 5x^4 - 8x, f'''(x) = 20x^3 - 8$

EXERCISE 2.8

1. Given $u = x^2$ and $v = \frac{u}{u-1}$, find $\frac{dv}{dx}$ at $x = 2$.

2. Given $u = x^3 - x$ and $v = \frac{u}{u^2+1}$ find $\frac{dv}{dx}$ at $x = 1$.

3. Find the points on the graph of $f(x) = x^3 - 3x + 2$ where the tangent line is horizontal.

4. Given $f(x) = g(x)h^3(x)$, $g(2) = -1$, $g'(2) = 3$, $h(2) = -2$, $h'(2) = 4$. Find $f'(2)$.

5. A tangent to the parabola $y = 3x^2 - 7x + 5$ is perpendicular to $x + 5y - 10 = 0$. Determine the equation of the tangent.

6. Find the point of tangency and the equation of the tangent line passing through the point $P(-1, -23)$ to the graph of the function $f(x) = 3 + 2x + 3x^2$.

Multiple Choice Circle the letter of the choice to the correct answer for the following 2 questions.

1. A function f is not differentiable at $x = a$ if:

- a. $f'(a)$ does not exist.
- b. $f(a)$ does not exist.
- c. the graph of f has a sharp corner, or cusp, at $x = a$.
- d. $f(x)$ is not continuous at $x = a$.
- e. all of the above.

2. If $f(x) = \frac{x-5}{3} - \frac{1}{x-3}$ then $f'(-1)$ is:

- a. $\frac{19}{48}$
- b. $\frac{4}{35}$
- c. $\frac{1}{2}$
- d. $\frac{29}{38}$
- e. $\frac{5}{3}$

3. Differentiate each of the following. Simplify your answers.

a. $y = \frac{2x^4}{\sqrt{x}} + \frac{8\pi^2}{\sqrt[4]{x}}$

b. $g(x) = \sqrt{2 - \sqrt{2x-1}}$

c. $f(x) = \frac{x + \frac{2}{x}}{x - \frac{1}{x}}$

d. $f(x) = \frac{\sqrt{\sqrt{x-1}}}{\sqrt{\sqrt{x+1}}}$

e. $y = \sqrt{\frac{x}{8}} - \frac{\pi}{8} + \pi x^2$

f. $f(x) = x\sqrt{x+2\sqrt{a}}$

g. $y = \left(\frac{x^2}{x-3}\right)^{-3}$

h. $g(x) = \frac{-2}{\sqrt{x^3 - 4x^2 + \pi}}$

4. Find $\frac{dy}{dx}$ for $y = (x^2 - 1)^2 (x^3 + 1)^3$. Express your answer in simplified factored form.

5. If $y = \frac{1-x}{1+x}$, find $\frac{d^2y}{dx^2}$. Express your answer in simplified factored form.

6. If $y = u^2 - 7u$, $u = -\sqrt{v+5}$ and $v = \frac{4}{(2-x)^2}$ find $\frac{dy}{dx}$ at $x = 3$.
7. If $y = -3 \left[x + (3x-5)^4 \right]^{-3}$, find $\frac{dy}{dx}$. Leave the answer in unsimplified form.
8. For $2(x^3-5)^3(1-x^2)^4$, Find $\left. \frac{dy}{dx} \right|_{x=2}$
9. A tangent to the parabola $y = 3x^2 - 7x + 5$ is perpendicular to $x + 5y - 10 = 0$. Determine the equation of the tangent.
10. Let $v = h(p(u))$ where $p(u) = u^{\frac{3}{2}} + 3u^3 - 5u$. If $h'(-1) = 5$, find $\frac{dv}{du}$ when $u = 1$.
-
11. If $y = ax^3 - 4a^2x - \frac{x}{2a-3b}$ find:
- a) $\frac{dy}{dx}$ b) $\frac{dy}{da}$
12. Consider the graph of $f(x) = x^3 - 9x$. The tangent has a y-intercept of 16. Determine the **exact** area of the triangle formed in the second quadrant by the tangent line and the axes.

Unit 2 Review

1. State each differentiation rule in both symbols and in words

- The Power Rule
- The Sum Rule
- The Product Rule
- The Chain Rule

2. Differentiate each of the following functions with respect to (w.r.t) x

a) $y = x^2 + x^3 - \frac{5}{\sqrt{x}} \quad \left[y' = \frac{5}{2x^{\frac{3}{2}}} + 3x^2 + 2x \right]$

b) $y = \frac{2x^3 - 7x}{4\sqrt{x}} \quad \left[y' = \frac{10x^2 - 7}{8\sqrt{x}} \right]$

c) $f(x) = 3\sqrt[3]{(2x^2 - 7)} + (5x^3 + 2)^3 \quad \left[f'(x) = x \left(45x(5x^3 + 2)^2 + \frac{4}{(2x^2 - 7)^{\frac{2}{3}}} \right) \right]$

d) $y = (4x^2 + 5)(x^3 - 27) \quad \left[\frac{dy}{dx} = x(20x^3 + 15x - 216) \right]$

e) $y = \frac{2}{x^2 - 7} + \frac{x^2 + 3}{2x^3} \quad \left[\frac{dy}{dx} = \frac{1}{x^2} - \frac{4x}{(x^2 - 7)^2} - \frac{3(x^2 + 3)}{2x^4} \right]$

f) $y = \sqrt{10 - 3x} \left(x^2 + \frac{3}{x^3} \right) \quad \left[-\frac{5(3x^6 - 8x^5 - 9x + 36)}{2x^4\sqrt{10 - 3x}} \right]$

g) $f(x) = \frac{(4x^3 + 5)^3}{\sqrt{3x + 5}} \quad \left[\frac{df(x)}{dx} = \frac{3(4x^3 + 5)^2(68x^3 + 120x^2 - 5)}{2(3x + 5)^{\frac{3}{2}}} \right]$

h) $y = \sqrt[3]{\frac{x^2 + 3x}{5x - 7}} \quad \left[\frac{dy}{dx} = \frac{5x^2 - 14x - 21}{3(x^2 + 3x)^{\frac{2}{3}}(5x - 7)^{\frac{4}{3}}} \right]$

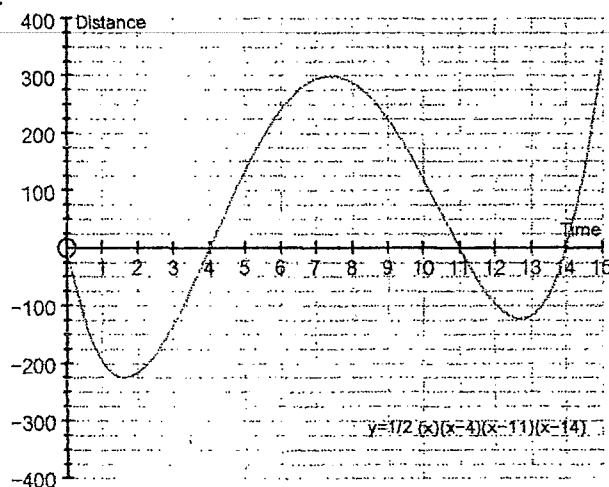
i) $f(x) = x^2(5x^2 + 3)^3(4x + 5)^{-2} \quad \left[f'(x) = \frac{10x(5x^2 + 3)^2(12x^3 + 20x^2 + 3)}{(4x + 5)^3} \right]$

~~$3x^2y^2 = x^3 + y^3 \quad \left[\frac{dy}{dx} = \frac{x^2 - 2xy^2}{y(2x^2 - y)} \right]$~~ omit

3. Suppose that $h(x) = f(x)g(x)$ and $F(x) = f(g(x))$, where $f(2) = 3$, $g(2) = 5$, $g'(2) = 4$, $f'(2) = -2$, and $f'(5) = 11$. Find $h'(2)$ and $F'(2)$ [$h'(2) = 2$, $F'(2) = 44$]

4. Find the points on the curve $y = \frac{1}{2x - 1}$ where the tangent line is perpendicular to the line $x - 2y = 1$ $[(0, -1), (1, 1)]$

5. Find the equation of the tangent line and normal line to the curve $y = x^3 - 3x^2 + 5$ at the point $(+1, 3)$. $[y = 3x + 6] \left[y = +\frac{1}{3}x + \frac{8}{3} \right]$
6. Find the point(s) on the parabola $y = -x^2 + 6x - 11$ where the tangent line at that point passes through the point $(2, 6)$. $[(-1, -18), (5, -6)]$
7. Find the value of a and b in $f(x) = ax^2 + bx$ given that $(-1, 5)$ is a point on the curve and the slope of the tangent at $x = 1$ is 7. $[a = 4, b = -1]$
8. Determine $\frac{dy}{dx}\bigg|_{x=1}$ if $y = \frac{1}{\sqrt{u-5}}$, $u = 3w^2 + 11$, $w = 2x + 3$ by using Leibniz Notation. $\left[-\frac{10}{243}\right]$
9. The equation of motion of a particle is given by $s(t) = 2x^3 - 33x^2 + 144x$.
a. Determine the velocity of the object when the acceleration is zero. $\left[-\frac{75 \text{ m}}{2 \text{ s}}\right]$
10. Find $\frac{d^3y}{dx^3}$ if $y = \frac{3}{(4-5x)^2} \left[y''' = \frac{9000}{(4-5x)^5} \right]$
11. Given the following Distance vs. Time graph, sketch graphs to represent Velocity vs. Time and Acceleration vs. Time.



12. Given the following Velocity vs. Time graph, sketch graphs to represent Distance vs. Time and Acceleration vs. Time.

